P3 — The Certificate as a Compliance Artifact

Certified Robustness Meets Airworthiness: Mapping ML Robustness Guarantees onto SORA Operational Volumes and ED-324 Learning Assurance

Research proposal, RobustUAVs.ai project · Roger Nick Anaedevha · drafted 2026-08-13
Follow-on to *From Wire to Flight* (NDSS submission)

Abstract

The parent manuscript already computes quantities that aviation regulators independently require, and does not say so loudly enough. Its corridor-margin family was chosen to bracket the JARUS SORA contingency volume; its certified tube radius is directly comparable to that volume's lateral extent; and the "operating region" qualifier on its Lipschitz hypothesis is, in the vocabulary of EASA's learning-assurance framework, an Operational Design Domain. This proposal formalises the correspondence, performs a gap analysis against the ED-324 learning-assurance objectives, and delivers an evidence generator that emits a SORA-shaped compliance pack from a certified run. It is a systematisation-and-tooling contribution, and is proposed as such.

1 The gap

Three developments have converged while the certified-robustness literature has looked the other way.

The operational-volume methodology is settled. Under the JARUS Specific Operations Risk Assessment, the operational volume comprises the flight geography plus a contingency volume whose minimum lateral extent is the sum of GNSS position accuracy, position-holding error, map error, a reaction-distance term, and the stopping distance of the contingency manoeuvre [11]. The parent manuscript's margin family $\{2, 5, 10, 20\}$ m was chosen to bracket exactly this, and its default values for the first three terms already total about 7 m.

The AI assurance framework is arriving. ED-324 / ARP6983 is EUROCAE WG-114 and SAE G-34's process standard for developing and certifying aeronautical products that implement AI [5, 4]. EASA's learning-assurance framework turns on a Concept of Operations, an Operational Domain, and an Operational Design Domain for the machine-learning constituent [7, 9], with the broader trajectory set out in the agency's AI roadmap [6].

The airspace framework is becoming mandatory. Network Remote ID is becoming a condition of participation in U-space [8, 1], which makes the connectivity assumptions of a cross-layer security argument regulatory rather than architectural.

Meanwhile the parent manuscript states its Lipschitz hypothesis as holding "over the operating region", and then enumerates that region: attitudes away from gimbal-lock singularities, airspeeds below stall, actuators off their saturation limits, no contact dynamics. **That is an ODD.** It is described in exactly those terms and the connection is never claimed.

2 The claim

A three-part contribution: a mapping, a gap analysis, and a generator.

2.1 The mapping

A correspondence between certified-robustness vocabulary — certified radius, trajectory tube, operating region, certified window, mission floor — and SORA/ED-324 vocabulary — contingency volume, ODD, operational domain, learning assurance objective. Where the correspondence is exact, prove it. Where it is not, name what is missing.

*The most likely honest finding is a negative one: **certified radii are per-input, while assurance objectives are per-operation**. A statement that a model is stable for every perturbation of norm at most R does not by itself discharge an objective phrased over a flight. The parent manuscript's conjunction of a spatial and a temporal predicate is a step across that divide, which is precisely why it is worth writing this paper — but the gap should be reported, not smoothed over. A systematisation whose finding is "it all lines up" would be worth very little.*

2.2 The gap analysis

For each ED-324 learning-assurance objective, classify what a Grönwall [10], randomized-smoothing [3], PAC-Bayes [12] or regret-based [2] certificate can do:

- **Discharge:** the certificate is sufficient evidence for the objective.
- **Evidence:** the certificate contributes but does not suffice.
- **Untouched:** the objective is about something certificates do not address — data governance, requirements traceability, human oversight.

The runtime-assurance reading already present in the parent manuscript — Simplex-style handover when the aircraft leaves the linearisable envelope [13] — is the natural bridge for objectives the certificate cannot discharge alone, and should be developed here rather than left as a remark.

2.3 The evidence generator

Emit a SORA Annex A-shaped evidence pack from a certified run: the ODD as the measured operating region, the contingency extent as the certified tube radius, the mission floor as the safety objective, and every figure carrying its provenance tag. This is what makes the paper more than a survey, and it is directly executable against the parent project's committed results.

3 What is reused, what is new

Reused unchanged	New
the entire compute side: engine, corridor family, provenance ledger	the vocabulary mapping and its proofs
the four-certificate architecture	the objective-by-objective gap analysis
the results already committed	the evidence generator over `results/`

This is the cheapest of the three proposals to execute and the only one with no measurement risk.

4 Relation to the parent manuscript's limitations

P3 closes none of the manuscript's stated limitations, and should not be sold as if it did. Its value is that it changes who the result is *for*: from a security audience that reads a certified floor as a bound, to a certification audience that reads it as evidence. That is a different contribution, not a smaller one, but the distinction should be explicit in the framing.

5 Platform integration

A **Compliance Pack** export: one action producing the evidence bundle for a chosen operating point, slotting into the existing export menu alongside the JSON, CSV, PNG and PDF exporters.

6 Risks

- **It is a systematisation paper and must be honest about that.** Its contribution is the mapping and the generator, not new theory. Pitch it to a venue that wants that, rather than overselling it at a top-tier security conference and being rejected for the right reason.
- **The standards move.** ED-324 is in preparation and the bibliography already records it as such [5]; the Commission's drone security package is in progress. Any claim keyed to a draft clause number will rot. Key claims to concepts — ODD, operational domain, contingency volume — which are stable, and cite clause numbers only in an appendix.
- **Regulatory over-claim.** Nothing in this paper certifies an aircraft. It maps a mathematical guarantee onto an evidence framework. The distinction must survive the abstract.

7 Venue and timeline

Target an IEEE S&P SoK track, ACM Computing Surveys, or the Digital Avionics Systems Conference — the last of which would put the work in front of the people who actually write these standards. About three months, and it can run in the gaps of P1 and P2.

References

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